

You must show your work to receive. Each problem is worth 4 pts unless stated otherwise.

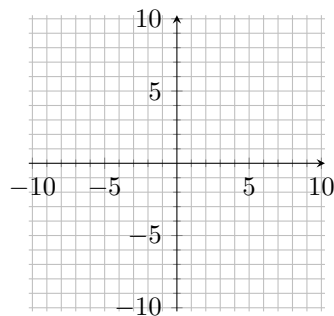
1. Use the function, $f(x) = 4^x$, to complete parts 1a–1d.

a. Graph $f(x)$. Label the y -int and at least one other point.

b. Find the domain of $f(x)$.

c. Find the range of $f(x)$.

d. Find the horizontal asymptote of $f(x)$.

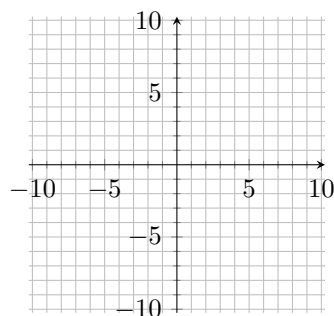


2. Use the function, $f(x) = \left(\frac{1}{4}\right)^x$, to complete parts 2a–2b.

a. Graph $f(x)$. Label the y -int and at least one other point.

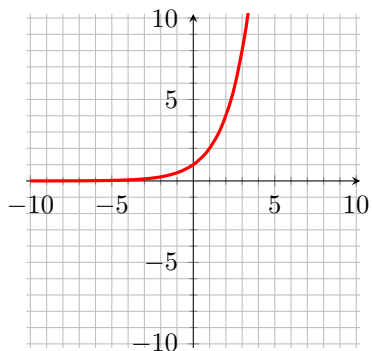
b. Find the horizontal asymptote of

$$g(x) = \left(\frac{1}{4}\right)^x + 2.$$

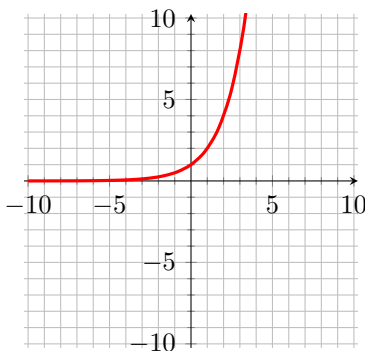


3. Use the graph of b^x in parts 3a–3c to graph the given transformation.

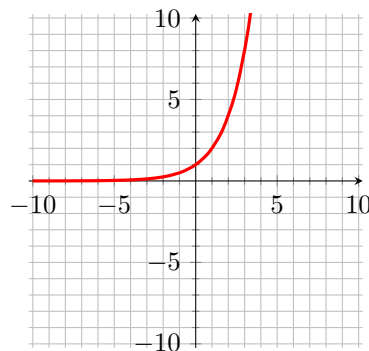
a. Graph $b^x + -2$



b. Graph b^{x++2}



c. Graph $b^{x++2} + -2$



Use the formulas, $A = P(1 + r)^t$, $A = P\left(1 + \frac{r}{n}\right)^{nt}$, or $A = Pe^{rt}$ to problems 4–6.

4. [6 points] Find the accumulated value of an investment of \$850 at 9% interest compounded quarterly after 20 years.

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5. [6 points] Bob buys a house for \$125,000. The value of the house grows at a rate of 4.5% compounded monthly. Find how much the house is worth after 15 years.
6. [6 points] Suppose Jane has \$1,000 to invest. Which investment yields the greater return over 10 years: 7% compounded monthly or 6.5% compounded continuously? Justify your answer.
7. Write the expression in its equivalent *exponential* form for parts 7a–7c.
- a. $\log_2 x = 100$ b. $\ln x = 2$ c. $\log_x 5 = y$
8. Write the expression in its equivalent *logarithmic* form for parts 8a–8c.
- a. $4^x = 64$ b. $x^{14} = 9$ c. $x^{-3} = \frac{1}{8}$
9. [3 points each] Evaluate or simplify the expression without using a calculator for parts 9a–9f.
- a. $\log_6 6$ c. $\log_2 2^{12}$ e. $\log_4 16$
- b. $e^{\ln 3}$ d. $4^{\log_4 10}$ f. $\log 10$
10. Find the domain of the logarithmic function for parts 10a–10b.
- a. $\log_{13} x$ b. $\ln(x - 3)$